

# Evaluating Multi-Agent Communication Topologies for Automated Code Review: A Pre-Registered Study

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**Abstract**—Large Language Models (LLMs) are increasingly used for automated code review, and recent multi-agent systems suggest that dividing work among specialized agents can improve reasoning on complex tasks. It remains unclear, however, whether the *communication topology* among agents—not merely their presence, nor the extra compute they consume—affects review quality, cost, and efficiency. This paper asks, under a pre-registered and compute-matched design, how communication topology affects the effectiveness and efficiency of automated code review. We compare four approaches under identical conditions: an Agentless single-LLM baseline, a compute-matched Generalists-3 control (the same generalist sampled three times and merged), a Hierarchical Authority architecture, and a Decentralized Peer Consensus architecture; the Generalists-3 control separates “more compute” from “specialization,” making our primary hypothesis testable. We have implemented a modular platform that reviews the same immutable pull-request snapshot with all four architectures through a shared import, review, validation, storage, and evaluation pipeline backed by a production LLM service, isolating topology as the only independent variable; the design, hypotheses, sample sizes, and analysis plan were pre-registered and prompts frozen before confirmatory data collection. Evaluation spans three complementary datasets: Qodo PR-Review-Bench (objective ground truth), SWE-PRBench (agreement with human reviewers), and an actively developed full-stack data portal used as an industrial case study, and reports both strict (file+line) and semantic (independent-judge) matching. The registered confirmatory campaign is complete: 99 Qodo and 50 SWE-PRBench PRs  $\times$  four architectures  $\times$  three runs, plus a pre-registered cross-family companion (Kimi K2.5, GLM 5). The primary specialization hypothesis is null—at matched compute, role specialists tie plain samples on recall ( $p=0.73$ )—and no multi-agent arm approaches the single-pass baseline on F1 (0.36–0.38 vs. 0.49, Holm  $p<0.001$ ): extra homogeneous agents buy recall (0.63 vs. 0.50) but drown it in false positives at  $3\text{--}9\times$  the cost, and the self-consistency rescue our pilot suggested does not replicate. The verification axis is different: findings corroborated by  $\geq 2$  independent model families reach precision 0.72 vs. 0.54 single-arm at equal-or-higher F1, and findings all three families agree on are real 89% of the time vs. 54% for one model repeating itself across three runs (gap CI [0.29, 0.41]; judge-invariant,  $\kappa=0.95$  across two independent pair judges). On the recall axis the diversity effect is small and saturating—unioning independent families adds only  $\sim 3$  points of coverage over temperature-matched sampling of a single model and plateaus within three families—so the multi-agent payoff is a precision effect, not a coverage one. In multi-agent code review, diversity pays on the verification axis, not the generation axis.

**Index Terms**—Large Language Models, Multi-Agent Systems, Automated Code Review, Software Engineering, Evaluation Methodology, LLM Verification.

## I. INTRODUCTION

Code review is a central quality-assurance practice: before code is merged, reviewers examine pull requests (PRs) for functional defects, security problems, maintainability issues, and architectural inconsistencies. It is also time-consuming and hard to scale, especially for full-stack systems spanning frontend, backend, databases, and deployment infrastructure. LLMs can summarize changes, flag bugs, and recommend fixes, and multi-agent frameworks such as ChatDev and MetaGPT suggest decomposing complex tasks among specialized agents [1], [2].

Existing work, however, usually evaluates complete agent systems rather than isolating the role of communication topology, and rarely separates the benefit of *communication* from the benefit of the additional *compute* that a multi-agent system spends. A centralized manager may behave very differently from a decentralized group in which agents discuss findings directly: centralized coordination may be faster and cheaper, while decentralized discussion may catch more cross-component issues. But either could simply be winning because it samples the model more times. We therefore study whether the structure of agent communication—holding the model, prompts, roles, and compute budget fixed—changes review outcomes:

**RQ:** How does the communication topology of a multi-agent system affect the effectiveness and efficiency of automated code review?

We refine this into three sub-questions:

- **RQ1 (Effectiveness).** Does topology change defect-detection quality (precision, recall, localization) and agreement with human reviewers?
- **RQ2 (Efficiency).** How does topology affect latency, token cost, and coordination overhead (LLM calls, inter-agent messages)?
- **RQ3 (Complexity interaction).** Does the effect of topology depend on PR complexity, e.g., localized versus cross-component changes?

We compare four architectures arranged as a ladder that changes one variable per rung: *Agentless*  $\rightarrow$  *Generalists-3* (+compute)  $\rightarrow$  *Hierarchical Authority* (+specialization)  $\rightarrow$  *Decentralized Peer Consensus* (+peer communication). Agentless is a strong single-LLM baseline, since simple non-agent

pipelines can be competitive on software-engineering tasks [3]. Generalists-3 runs the identical generalist prompt three times in parallel and merges the results with the shared synthesizer: it holds the sampling budget equal to the multi-agent arms but adds no roles and no communication, so it is the compute-matched control that makes the primary (specialization) hypothesis testable. Hierarchical and Consensus share specialist roles but differ in how agents communicate.

Our contributions are:

- A controlled design—pre-registered, with prompts frozen before confirmatory data collection—and an implemented, reproducible platform in which review architecture is the only independent variable, including a compute-matched generalist control.
- A three-dataset evaluation framework combining objective benchmark correctness, human-reviewer agreement, and an industrial case study, with both strict and semantic matching.
- An industrial verification methodology that corroborates findings without authoritative ground truth.
- Confirmatory results: a pre-registered null for role specialization and for homogeneous multi-agent F1 gains, a confirmatory refutation of our own pilot’s self-consistency rescue, and a confirmed cross-family corroboration hypothesis—agreement between independent model families is a quantified, judge-invariant precision instrument (89% vs. 54% at full agreement)—plus a three-judge audit showing benchmark ground truth is materially incomplete.

## II. RELATED WORK

### A. LLM-Based Software Engineering

LLMs have been applied to code generation, program repair, bug localization, and test generation. SWE-bench benchmarks real GitHub issues and PRs and shows the difficulty of resolving realistic problems with language models [5]. Agentless challenges the assumption that complex agent systems are always necessary, using a simple localize–repair–validate pipeline [3]; its strong performance makes it an essential baseline for judging whether multi-agent communication adds measurable value.

### B. Multi-Agent Software Engineering

Multi-agent systems assign roles to multiple LLM agents. ChatDev models development as communication among design, coding, and testing agents [1]; MetaGPT adds structured workflows and role-based collaboration to reduce inconsistency [2]. Surveys argue the area is promising but under-evaluated empirically [6], and other work finds that agent systems fail through coordination overhead, cascading errors, and weak verification [7]. These failure modes are precisely properties of communication structure, motivating our controlled comparison of topologies rather than of whole systems. Concurrent work sharpens the negative case: Pappu et al. show LLM teams underperform their strongest member by up to 41% through “integrative compromise”—averaging expert

and non-expert views rather than weighting expertise [13]—and Li et al. find that mixing different LLMs rarely beats self-ensembling the best single model (Self-MoA) [14]. Our compute-matched, pre-registered design tests these dynamics in code review specifically; our confirmatory nulls replicate them under a frozen-model discipline, and our cross-family result locates where mixing *does* pay.

### C. Evaluation Without Ground Truth

Objective benchmarks require labeled defects, which real industrial repositories rarely provide. Using a separate LLM as an impartial judge is an increasingly common way to assess outputs at scale [8]. We adopt LLM-as-a-judge as one corroboration signal—not a source of ground truth—and combine it with cross-architecture agreement and conventional static analysis to triangulate the credibility of findings on a live repository.

### D. Verifying LLM Outputs

A parallel line of work asks not how to *generate* candidate outputs but how to *verify* them. Cobbe et al. show that a trained verifier can outperform a much larger generator by re-ranking sampled solutions [9]. For LLM-as-a-judge specifically, Verga et al. find that a panel of smaller diverse judges beats a single large judge [10], while Panickssery et al. document a self-preference bias in which evaluators favor their own generations [11]—directly motivating our non-circular design in which generator, matching judge, and verifier are three different model families. Kwok et al. frame verification as a general-purpose capability with three scaling axes—scoring granularity, repeated evaluation, and criteria decomposition [12]—the structure our strongest verifier (V3 below) adapts. Closest to our confirmed hypothesis, Zietsman conjectures a *correlated-error* failure mode for AI code review—same-family generator and reviewer share blind spots, so “combining does not reduce error, it consolidates it”—and contrasts same- vs. cross-family reviewers on a small contrived corpus, self-described as directional and unpowered [15]. We provide what that argument lacks: a pre-registered, statistically powered test against an external injected-defect ground truth, operationalizing cross-family agreement as a precision instrument (Section VI-E).

## III. RESEARCH DESIGN

### A. Variables

The independent variable is communication topology across four architectures: (1) *Agentless*: one LLM reviews the PR without agent communication; (2) *Generalists-3*: the same generalist prompt sampled three times in parallel and merged by the shared synthesizer—no roles, no communication—serving as the compute-matched control; (3) *Hierarchical Authority*: a manager assigns work to specialists and synthesizes their findings; (4) *Decentralized Peer Consensus*: specialists review independently, exchange findings, and vote to converge. Controlled variables (model, prompts, roles, sampling budget, PR snapshot) are held fixed; dependent variables capture effectiveness and efficiency (Table I).

TABLE I  
RESEARCH VARIABLES

| Type          | Description                                         |
|---------------|-----------------------------------------------------|
| Independent   | Communication topology (4 architectures)            |
| Controlled    | Model, prompts, roles, sampling budget, PR snapshot |
| Effectiveness | Precision, recall, F1, localization; corroboration  |
| Efficiency    | Latency, token cost, LLM calls, message count       |

### B. Hypotheses

The design, hypotheses, sample sizes, and analysis plan were pre-registered on OSF (embargoed) and prompts frozen (tag `prompt-freeze-v1`) before any confirmatory data collection. The registered contrasts are:

- **H-specialization (primary).** Role-specialized teams (Hierarchical, Consensus) outperform the compute-matched Generalists-3 control on defect-detection quality (RQ1). This is the confirmatory hypothesis; the Generalists-3 control is what makes it distinguishable from a mere compute effect.
- **H-compute (secondary).** Generalists-3 outperforms Agentless, isolating the value of additional sampling independent of structure (RQ1).
- **H-communication (secondary).** Peer communication (Consensus) outperforms coordinator-mediated aggregation (Hierarchical), especially on recall for cross-component PRs (RQ1, RQ3).
- **H-efficiency (secondary).** Latency, token cost, and coordination overhead rise monotonically along the ladder Agentless  $\rightarrow$  Generalists-3  $\rightarrow$  Hierarchical  $\rightarrow$  Consensus (RQ2).
- **H-complexity (secondary).** The effect of topology grows with PR complexity (RQ3).

Two secondary confirmatory hypotheses were added by a pre-data OSF amendment (submitted before any confirmatory collection), motivated by the exploratory pilots: **H-verify**—a self-consistency filter (keep findings recurring in  $\geq 2$  of 3 runs) lifts each multi-agent arm’s semantic F1 to within 0.02 of the Agentless baseline; and **H-hetero-precision**—findings corroborated by  $\geq 2$  of three independent model families {Claude Haiku 4.5, Kimi K2.5, GLM 5} achieve precision  $\geq 10$  points above the mean single-arm precision at equal-or-higher F1, and the golden-match rate of findings agreed by all three families exceeds the frozen model’s own three-run recurrence by  $\geq 10$  points, claimed only if the CI excludes a  $< 5$ -point gap. The companion families were selected by a pre-specified parity gate (single-arm semantic F1  $\geq 0.85 \times$  the best family’s), and the  $\leq 21$  pilot PRs used for that screening are excluded from the H-hetero-precision test set. A model-scale arm (Claude Sonnet 4.5 replacing Haiku 4.5) is exploratory, used to check that the architecture ordering is model-robust rather than an artifact of a single generator.

## IV. PLATFORM AND METHODOLOGY

### A. Experimental Platform

We implemented a modular platform that drives every architecture through the same pipeline: PR import produces an immutable snapshot; the architecture under test reviews that snapshot; a validation stage normalizes raw model output into structured findings; results are stored; and an evaluation engine computes metrics. A campaign runner imports each PR once and reviews the shared snapshot with all four architectures, so topology is the only variable (fairness policy). All arms run **Claude Haiku 4.5** as the frozen system under test (with **Claude Sonnet 4.5** as a robustness arm), with temperature and prompts frozen; reviews run on Amazon Bedrock. Semantic matching and verification use non-Anthropic model families—**Llama 3.3 70B** as the finding $\rightarrow$ ground-truth judge, **Amazon Nova Pro** as the cross-family finding $\leftrightarrow$ finding pair judge, and **DeepSeek V3.2** as verifier and second judge—so no model family scores its own family’s output, and each judging role is cross-checked by a second independent family.

### B. Architectures

Figure 1 contrasts the four architectures. Agentless reviews the diff in a single pass. Generalists-3 runs that same generalist review three times in parallel and merges the outputs through the shared synthesizer, matching the multi-agent sampling budget without adding roles or communication. Hierarchical Authority has a coordinator (manager) direct frontend, back-end, and database specialists, then deterministically merges their findings. Decentralized Peer Consensus uses the same specialists, which first review independently, then exchange findings and vote to reach a shared set. The multi-agent architectures share specialist implementations and prompts, differing only in coordination and message flow.

### C. Datasets

Three datasets provide complementary evidence: *Qodo PR-Review-Bench* (PRs with injected defects and explicit ground truth  $\rightarrow$  precision, recall, F1, localization); *SWE-PRBench* (real PRs with human review comments, interpreted as agreement with reviewers rather than absolute correctness); and an *industrial case study*—an actively developed full-stack data portal (React frontend, backend APIs, database logic, authentication, cloud deployment)—reviewed on real PRs for ecological validity but without authoritative ground truth.

### D. Evaluation Metrics

Metrics are read according to the evidence each dataset offers (Table II). For the industrial case study, correctness cannot be measured directly, so we do not report precision/recall/F1. Instead, *industrial verification* corroborates findings via: (i) Cross-Architecture Agreement—the fraction of an architecture’s findings independently found by another architecture on the same PR (matched on file, line, category, and title similarity); (ii) Static-Analysis Agreement—the fraction coinciding with issues from conventional static-analysis tools; (iii) LLM-Judge Validation—the fraction a separate,

## Comparison of Multi-Agent Communication Topologies for Automated Code Review

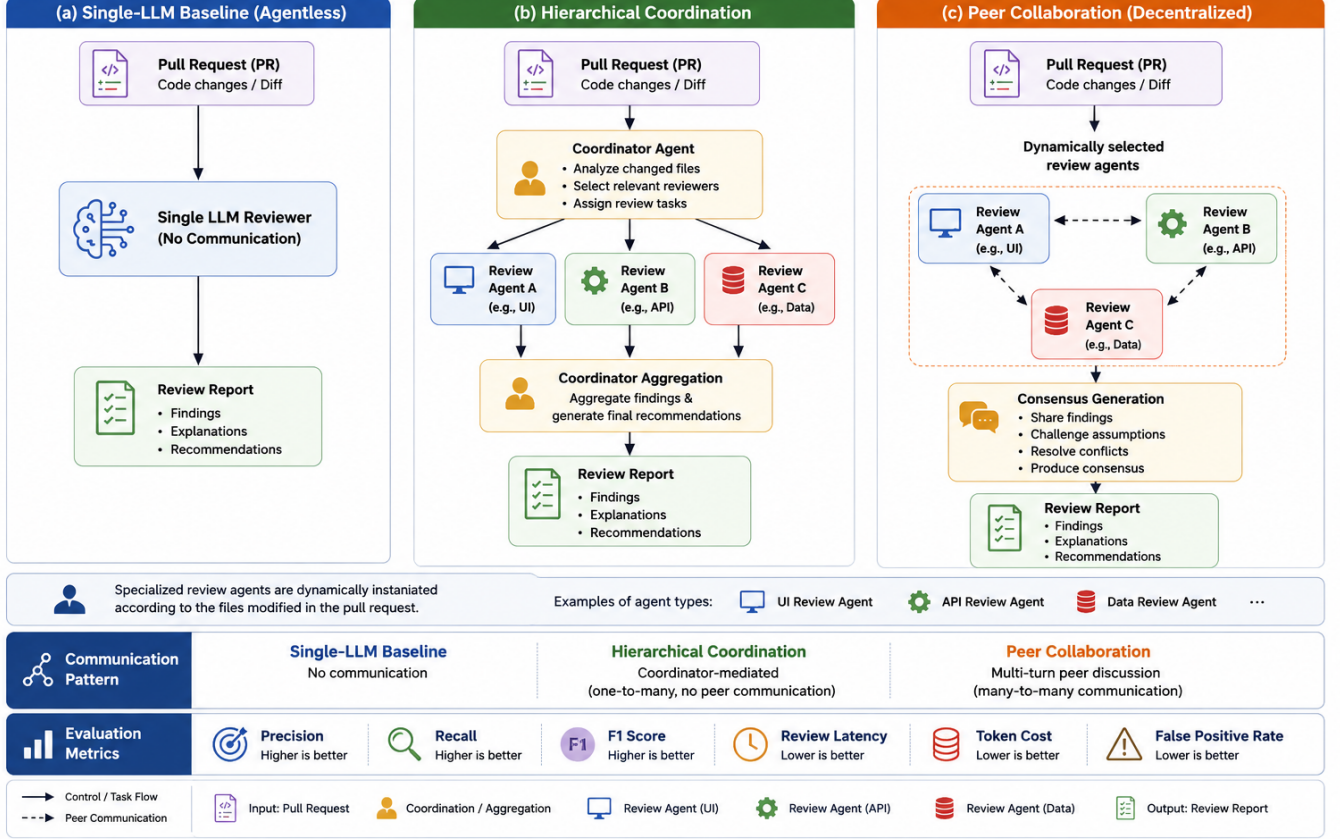


Fig. 1. The three communication topologies compared. (a) *Agentless*: a single LLM reviews the pull request with no agent communication. (b) *Hierarchical Authority*: a coordinator assigns the diff to specialist agents (e.g., UI, API, data) and aggregates their findings—a one-to-many, coordinator-mediated pattern with no peer communication. (c) *Decentralized Peer Consensus*: dynamically selected specialists review independently and then discuss directly (many-to-many) to resolve conflicts and produce a consensus report. Solid arrows denote control/task flow; dashed arrows denote peer communication. The compute-matched *Generalists-3* control (Section III-A) is Agentless sampled three times in parallel and merged—no roles, no communication—so it introduces no new communication topology and is described in the text rather than shown as a separate panel.

impartial LLM classifies as valid given the diff; (iv) Later-Fix Rate—the fraction whose location is modified in later commits; and (v) Evidence Score, a supporting heuristic over severity, confidence, and finding volume that reflects review strength, not correctness. Operational metrics (latency, tokens, cost, LLM calls, messages) are collected identically across all datasets.

Formally, let  $F$  be an architecture’s validated findings on a PR and  $A_x$  the findings of architecture  $x$ . Writing  $m(f, S) = 1$  when a finding  $f$  matches some element of set  $S$  (same file, a small line window, and matching category or title), the

corroboration signals are

$$CAA_x = \frac{1}{|A_x|} \sum_{f \in A_x} \mathbf{1}[\exists y \neq x : m(f, A_y)], \quad (1)$$

$$SAA = \frac{1}{|F|} \sum_{f \in F} \mathbf{1}[m(f, S_{\text{static}})], \quad (2)$$

$$LJV = \frac{1}{|F|} |\{f \in F : \text{judge}(f) = \text{valid}\}|, \quad (3)$$

$$LFR = \frac{1}{|F|} |\{f \in F : \text{loc}(f) \text{ changed later}\}|, \quad (4)$$

where  $S_{\text{static}}$  is the set of static-analysis issues and  $\text{judge}(\cdot)$  is an independent LLM verdict. Each value lies in  $[0, 1]$  and is reported only when computable—for example, cross-architecture agreement requires at least two architectures and a non-empty finding set.

Two matching refinements support fair comparison across arms. *Semantic matching*: strict file+line matching under-credits near-miss findings, so an independent LLM judge makes a binary same-issue decision that rescues same-file

TABLE II  
EVALUATION METRICS BY DATASET

| Metric                       | Qodo | SWE-PRBench | RAP Portal |
|------------------------------|------|-------------|------------|
| Precision / Recall / F1      | ✓    | ○           | —          |
| Localization Accuracy        | ✓    | —           | —          |
| Human Review Agreement       | —    | ✓           | —          |
| Cross-Architecture Agreement | —    | —           | ✓          |
| Static-Analysis Agreement    | —    | —           | ✓          |
| LLM-Judge Validation         | —    | —           | ✓          |
| Later-Fix Rate               | —    | —           | ✓          |
| Evidence Score (heuristic)   | —    | —           | ✓          |
| Latency / Cost / Tokens      | ✓    | ✓           | ✓          |
| LLM Calls / Message Count    | ✓    | ✓           | ✓          |

✓: reported; ○: interpreted as human agreement; —: not applicable.

findings without exact line overlap; we report both strict and semantic metrics throughout. The judge is effectively threshold-insensitive across the 0.5–0.9 range, removing a hyperparameter-sensitivity concern. *Lenient per-finding validation*: the review envelope stays strict, but individual malformed findings are dropped and logged rather than failing the entire review—this removes a survivorship bias that previously penalized the single-call arm, where one malformed finding could kill a whole review.

#### E. System Implementation

The platform is organized as a set of independent stages connected by explicit data contracts, which keeps the experiment reproducible and makes each stage separately testable. Import converts a PR into an immutable snapshot keyed by a content-derived identifier, so re-running an experiment reuses the exact same input. Review invokes the architecture under test, which returns raw model output plus execution telemetry (latency, tokens, cost, LLM calls, messages). Validation parses the raw output against a fixed schema and normalizes it into structured findings without inventing data, so malformed output degrades gracefully rather than corrupting metrics. Storage persists raw and validated results and individual findings; evaluation computes the metrics of Section IV-D; and export emits per-experiment and comparison rows as CSV and JSON for analysis.

Two properties support internal validity. First, the evaluation engine is deterministic and side-effect free: it reads only stored artifacts and never calls the model, so metrics are reproducible from saved results—and, since all intermediate artifacts are persisted, verifier ablations can be run entirely on cached findings with zero new generation. Second, the campaign runner maintains a manifest of every (PR, architecture, run) with status, attempts, and reproducibility metadata (model, prompt, and workflow versions), enabling resume-after-failure and auditing; a retry policy re-attempts only transient infrastructure errors, while validation and parsing errors are terminal. The industrial-verification signals are additive: they extend the research-evidence metrics without altering the benchmark correctness path, so the Qodo and SWE-PRBench pipelines are unaffected by the case-study methodology.

## V. THE CONFIRMATORY CAMPAIGN

### A. Campaign Design

The primary experiment is a full campaign in which every PR is reviewed by all four architectures over the shared snapshot. Because the pipeline reuses one import per PR, all architectures see byte-identical input, so any outcome difference is attributable to topology (or, for Generalists-3, to compute). Each run records reproducibility metadata (model version, prompt version, workflow version) and a deterministic manifest so runs can be resumed and audited; a retry policy re-attempts only transient infrastructure failures, while validation and parsing errors are terminal.

### B. Datasets and Sample Sizes

The registered design fixes **100 PRs from Qodo, 50 from SWE-PRBench, and 15 from the industrial portal, each** reviewed under all **four architectures with three runs each**. The benchmark portion is now complete: 99 Qodo PRs (one PR was excluded campaign-wide under the pre-specified envelope-failure rule after persistently malformed responses on one arm) and all 50 SWE-PRBench PRs— $149 \times 4 \times 3 = 1,788$  reviews of the frozen Haiku 4.5 SUT—plus the pre-registered companion generation: agentless reviews by Kimi K2.5 and GLM 5 (three runs each, unchanged frozen prompt) over the same Qodo PRs, giving the cross-family analysis its 80-PR disjoint test set after the pilot exclusion. The industrial-portal arm is deferred (its static-analysis and later-fix corroboration producers are still under construction) and is not part of the results below. PRs are stratified by type—frontend-only, backend-only, database-only, and cross-component—so that RQ3 can be tested against complexity. A power analysis on the pilot-measured paired recall standard deviation ( $\approx 0.13$ ) indicates that  $N=100$  detects a  $\approx 3.6$ -point recall difference, or bounds a null within  $\pm 2.5$  points. Benchmark defect categories include missing authorization checks, incorrect API contracts, frontend state bugs, query inefficiencies, missing validation, error-handling problems, and cross-component integration failures.

### C. Procedure and Analysis

Each PR flows through import  $\rightarrow$  review ( $\times 4$ )  $\rightarrow$  validation  $\rightarrow$  storage  $\rightarrow$  evaluation  $\rightarrow$  export. Effectiveness is measured objectively on Qodo (precision, recall, F1, localization), as human agreement on SWE-PRBench, and by corroboration on the industrial portal. Because SWE-PRBench golden comments are location-less, they are evaluated by semantic coverage (judge: “same underlying issue?”) rather than file+line, mirroring the benchmark’s own methodology. Efficiency is measured everywhere (latency, tokens, LLM calls, messages).

Because the same four architectures review every PR, observations are paired, and we compare architectures with paired non-parametric tests (Wilcoxon signed-rank for two-way contrasts, Friedman across all four) rather than assume normality, reporting effect sizes alongside  $p$ -values and correcting for multiple comparisons. To address RQ3 we stratify PRs by type and test for an interaction between topology

and complexity; H-communication predicts that Consensus gains most on cross-component PRs. Cost is reported both in absolute terms and as a cost-per-true-positive ratio, so that any accuracy gain from richer coordination is weighed against its overhead. All prompts and the model are frozen across architectures to preserve internal validity, and every run stores its reproducibility metadata so the analysis can be regenerated from saved artifacts.

## VI. CONFIRMATORY RESULTS

The registered campaign is complete over both benchmark datasets and analyzed with the registered paired framework: the PR is the unit of analysis, each (PR, arm) value is the mean of three runs, and contrasts use paired Wilcoxon signed-rank tests, Cliff’s  $\delta$ , seeded percentile-bootstrap CIs, and Holm–Bonferroni correction within each hypothesis family. All generation ran under the frozen prompts and model; every raw output and judge score is persisted, so each number below replays deterministically from saved artifacts with zero new LLM calls. The earlier exploratory results (the  $N=20$  pilot, the verifier-strength ablation, and the first completeness audit) are revisited only where the confirmatory data confirms or overturns them.

### A. Generation axis: more agents buy recall, never quality

Table III gives per-arm macro means over the 99 Qodo PRs. The **primary hypothesis (H-specialization) is null**: at matched compute, Hierarchical ties Generalists-3 on recall (0.624 vs. 0.626;  $\hat{\Delta}=0.000$ ,  $p=0.73$ ,  $\delta=-0.005$ , CI  $[-0.07, 0.06]$ ) at comparable precision (0.294 vs. 0.270,  $p=0.10$ ). Role specialization contributes nothing beyond sampling—consistent with concurrent evidence that LLM teams average rather than exploit expertise [13]. Nor does compute convert into quality: Agentless dominates every multi-agent arm on semantic F1 (0.487 vs. 0.357–0.378; all Holm  $p<0.001$ ,  $\delta$  0.30–0.36). What extra compute buys is recall: Generalists-3 and Hierarchical reach 0.62–0.63 vs. 0.503 (Holm  $p<0.001$ ), though Consensus—the most expensive arm—does not (0.536, Holm  $p=0.12$ ), leaving H-communication unsupported. Efficiency makes the trade stark: one golden-confirmed finding costs 0.34 LLM calls under Agentless and 1.76 under Consensus (pooled, 80-PR subset)—a  $5\times$  premium for a lower-F1 review. An exploratory Sonnet 4.5 pilot preserves the ordering (single-pass 0.57 vs. 0.39–0.44), suggesting the ranking is not a Haiku artifact, though the robustness arm has not run at confirmatory scale.

On SWE-PRBench ( $N=50$ ; semantic coverage of human review comments, matching the benchmark’s location-less methodology) the same shape replicates in natural data: Generalists-3 raises coverage (0.68 vs. 0.51, Holm  $p=0.001$ ; Hierarchical 0.63, Holm  $p=0.05$ ), but F1 is a four-way tie (Agentless 0.325 vs. 0.329–0.357; all n.s.,  $|\delta| \leq 0.06$ ).

### B. Model capability is not a defect-type-targeted lever

A registered amendment tested whether a stronger model’s advantage concentrates on reasoning-limited functional bugs

TABLE III  
CONFIRMATORY PER-ARM MACRO MEANS, SEMANTIC MATCHING ( $N=99$  QODO PRs; PER-PR MEAN OF 3 RUNS). FDR =  $1-\text{PRECISION}$ .

| Architecture  | Precision    | Recall       | F1           | FDR          |
|---------------|--------------|--------------|--------------|--------------|
| Agentless     | <b>0.525</b> | 0.503        | <b>0.487</b> | <b>0.475</b> |
| Generalists-3 | 0.270        | <b>0.626</b> | 0.357        | 0.730        |
| Hierarchical  | 0.294        | 0.624        | 0.378        | 0.706        |
| Consensus     | 0.322        | 0.536        | 0.369        | 0.678        |

(a capability  $\times$  defect-type interaction). On the 80-PR disjoint remainder, agentless-ungrounded, the primary difference-in-differences is **null**:  $\text{DiD} = \Delta\text{recall}_{\text{func}} - \Delta\text{recall}_{\text{rule}} = -1.8$  points (CI  $[-8.6, 5.5]$ ,  $p=0.68$ ). Sonnet 4.5 improves both types modestly over Haiku (functional 0.70 $\rightarrow$ 0.74, rule 0.32 $\rightarrow$ 0.37) at equal or better precision—a broad, undirected lift, not a targeted one. The pilot’s +10-point functional-only interaction reproduces only on the two convention-dense pilot repositories and vanishes elsewhere. Capability/scale buys uniform recall, not a defect-class-specific capability.

### C. No generation-side structure beats compute-matched sampling

The specialization null (Sec. VI-A) is not specific to *roles*; two further ways to structure the generation side also fail (exploratory). **Module decomposition**: partitioning each PR by code module, reviewing every module with its own agent, and unioning beats a single whole-PR review by +6.4 points of recall ( $p=0.016$ ,  $N=50$ )—but that is the  $\sim 2.6\times$  extra compute, not the decomposition: at a matched call budget, module-union *trails* plain multi-sampling on the multi-module-GT subset (0.54 vs. 0.57;  $\hat{\Delta}=-0.029$ ,  $p=0.44$ ,  $n=24$ ). **Error-type specialists**: prompting one model to report *only* convention issues and another *only* functional bugs, then unioning, is worse—focus *reduces* per-type recall (convention specialist 24% of rule violations vs. the generalist’s 32%; functional 59% vs. 68%) because “report only  $X$ ” suppresses output (5.3 findings/PR vs. 16.1), and the team reaches recall 0.47, below the generalist (0.50) and the same-budget baselines (multi-sample 0.58, cross-family 0.60). Prompt “personas” on a single model are a negative decorrelation source. Type specialization pays only when it changes the detection *mechanism* (a deterministic linter, execution, or an independent model family), not the prompt.

### D. H-verify: the pilot’s self-consistency rescue does not replicate

The exploratory verifier ablation had suggested that a free self-consistency filter (V1: keep findings recurring in  $\geq k$  of 3 runs) rescues the multi-agent arms to baseline parity, and H-verify registered exactly that prediction (parity band: within 0.02 of Agentless F1). **Confirmatorily, it fails.** At  $k=2$ , verified Generalists-3 reaches 0.447 vs. baseline 0.487 ( $\hat{\Delta}=-0.042$ , CI  $[-0.070, 0.000]$ ; inconclusive), while Hierarchical (0.378;  $\hat{\Delta}=-0.127$ , CI  $[-0.179, -0.083]$ ) and Consensus (0.373;  $\hat{\Delta}=-0.114$ , CI  $[-0.167, -0.068]$ ) fall entirely

below the band;  $k=3$  fails for every arm. The pilot’s rescue was a small-batch artifact. The mechanism becomes visible in the corroboration analysis below: a model that agrees with itself is mostly repeating its confident errors, so same-model recurrence filters little. The ablation’s remaining findings stand and were not registered as rescues: a binary frontier-model judge is a rubber stamp (approving  $\sim 83\%$  of everything it is shown), and rubric-structured scoring discriminates weakly (AUC 0.68) but converts into no F1 gain—verifier *structure*, not judge intelligence, remains the operative variable; the confirmatory data relocates the effective structure from self-recurrence to cross-family agreement.

*E. H-hetero-precision: cross-family agreement is a precision instrument (confirmed)*

The pre-registered companion generated agentless reviews with two independent frontier families—Kimi K2.5 and GLM 5, both clearing the parity gate (observed ratios 0.95–1.00)—over the same PRs with the unchanged frozen prompt. Cross-model finding $\leftrightarrow$ finding matching uses a fourth family (Amazon Nova Pro) as pair judge, keeping the judging chain non-circular. On the registered 80-PR disjoint remainder, **both sub-claims hold**:

- **Precision at  $\geq 2$ -family corroboration.** Corroborated findings reach pooled precision 0.697 vs. 0.542 mean single-arm; per-PR paired ( $n=79$ ): 0.716 vs. 0.575,  $\bar{\Delta}=+0.141$ , CI [0.094, 0.189],  $p<10^{-4}$ ,  $\delta=0.364$ —at equal-or-higher F1 (0.511 vs. 0.487, CI [−0.004, 0.051]).
- **All-three-family agreement vs. self-recurrence.** Findings all three families independently produce are golden-matched **89%** of the time ( $n=137$  clusters), vs. **54%** ( $n=391$ ) for findings the frozen model repeats across all three of its own runs: gap +0.348, CI [0.286, 0.412]—far beyond the registered +0.10 threshold, with the CI excluding a  $<0.05$  gap.

Table IV stratifies by corroboration depth: cross-family agreement climbs steeply (28% $\rightarrow$ 51% $\rightarrow$ 89%) where same-model recurrence stays flat before saturating at its error ceiling (14% $\rightarrow$ 17% $\rightarrow$ 54%). The equal-depth contrasts (+34 and +35 points) are the correlated-error mechanism made visible: a second run of the same model contributes almost no independent evidence, while a second independent family does. The instrument is judge-invariant: re-judging all 8,061 candidate pairs with an independent second pair judge (DeepSeek V3.2) agrees with Nova at  $\kappa=0.952$  (raw 97.9%, Pearson  $r=0.949$ ) and reproduces the headline rows, and the result is insensitive to the pair threshold across 0.5–0.9.

*F. The recall axis is a decorrelation ladder (temperature-dominated)*

The confirmed cross-family result above is a *precision* instrument; on the orthogonal *recall* axis the cross-family gain is real but small once compute is matched fairly (exploratory). Table V unions  $K$  agentless reviews two ways on the same 97 Qodo PRs:  $K$  temperature-diverse samples of one model (Haiku,  $T=0.7$ ) vs.  $K$  distinct families. Temperature-sampling

TABLE IV  
GOLDEN-MATCH RATE BY CORROBORATION DEPTH (SEMANTIC MATCHING; 80-PR REGISTERED SET). SAME-MODEL = FROZEN HAIKU ACROSS ITS 3 RUNS; CROSS-FAMILY = ONE RUN EACH OF HAIKU, KIMI K2.5, GLM 5.

| Sources agreeing           | 1         | 2         | 3                |
|----------------------------|-----------|-----------|------------------|
| Same model $\times 3$ runs | 14% (35)  | 17% (30)  | 54% (391)        |
| Cross-family $\times 3$    | 28% (419) | 51% (114) | <b>89% (137)</b> |

Cell = golden-match rate ( $n$  clusters).

TABLE V  
UNION RECALL BY NUMBER OF DRAWS  $K$  (SEMANTIC MATCHING, 97 QODO PRs). HOMO = ONE MODEL (HAIKU) AT  $T=0.7$ ; HETERO = HAIKU PLUS ONE DISTINCT FAMILY PER RUNG (KIMI, GLM, DEEPSEEK, NOVA, LLAMA 4, PALMYRA).

| $K$               | 1    | 2    | 3    | 4    | 5    | 6    | 7    |
|-------------------|------|------|------|------|------|------|------|
| Homo ( $T=0.7$ )  | 0.49 | 0.58 | 0.61 | —    | —    | —    | —    |
| Hetero (families) | 0.51 | 0.61 | 0.64 | 0.65 | 0.65 | 0.66 | 0.66 |

Diversity bonus (hetero − homo) = +0.03 at  $K=2, 3$ . Family order affects per-step marginals, not the  $K=\max$  total.

a single model already reaches 0.61 recall at  $K=3$ ; adding whole families lifts it only to 0.64—a +3-point diversity bonus that saturates by  $\sim 3$ –4 families (ceiling  $\sim 0.65$ –0.66; findings keep rising with flat recall and eroding precision). Most union-recall gain is a single model recovering issues it can find but does not surface in one pass (self-coverage); cross-family adds only the thin complement it cannot reach at any temperature. A same-model union at  $T=0$  (near-identical runs) gives essentially nothing (+0.9 points), so comparing cross-family against a  $T=0$  baseline overstates the bonus; the fair, temperature-matched figure is +3 points.

*G. Where the signal lives*

Stratifying the 80-PR set by Qodo’s two injected defect types shows the cross-family signal is carried by the *hard* class. Of the in-set functional bugs (217), 80%/61%/43% are caught by  $\geq 1/2/3$  independent families, versus 45%/26%/18% of the project-specific rule violations (220): a generic reviewer cannot know repository-local conventions, but universal functional defects are exactly where independent families converge. The  $\geq 1$ -family coverage figure is a union over three families and is therefore temperature-dominated (a single model sampled three times reaches most of it, Table V); the cross-family contribution is carried by the agreement-depth (precision) result above, not by coverage volume. The  $\geq 2$ -family “trustworthy” set is 52% functional true positives, 20% rule true positives, and 28% golden-unmatched. False-positive anatomy adds two correctives: 21–32% of golden-unmatched findings are near-misses within ten lines of a real defect (localization slop, not hallucination), and 91% of false positives are low/medium severity—the arms’ high-severity findings are almost always real.



## H. The diff-scoped ceiling and a division of labour

Pushing decorrelation to its limit—the union of *every* configuration we ran (eleven sources: seven model families, a conditioned “find-new” resampling arm, and a temperature sweep)—reaches only 0.83 recall on the 97-PR set, and leave-one-out shows no single source contributes more than  $\sim 2$  points, so the ceiling is robust (exploratory). The residue is not random. Functional bugs are 89% robustly reachable (covered by  $\geq 2$  sources), whereas 36% of the project-specific rule violations are reached by *no* configuration: reasoning errors decorrelate across models, but a mechanical-convention blind spot is *correlated*, so eleven draws miss the same violations. Categorising the unreachable core,  $\sim 65\%$  are mechanical conventions a deterministic linter targets and  $\sim 15\%$  are a hard functional core (cross-file renames, dynamic imports, races, harmful deletions) invisible to a diff-scoped reviewer. Wiring in a linter confirms the division of labour: on the two convention-authored pilot repositories, a single deterministic checker out-recalls the entire eleven-model union on exactly the rules it targets (0.54 vs. 0.46), and their hybrid reaches 0.61—the linter recovering  $\sim 27\%$  of the rule violations no amount of sampling reached. Multi-agent review thus has three separable jobs, each with its own tool: *lint* the mechanical conventions, *union-sample across families* for functional bugs, and reserve *agentic retrieval or execution* for the hard functional core. No precision back-end changes this picture: neither cross-family agreement (Section VI-E) nor MAV-style decomposed aspect verification lifts  $F_1$  above a plain independent union—filters and verifiers trade recall for precision but do not move the frontier.

## I. Ground-truth completeness: direction robust, magnitude definitional

The pilot audit estimated the golden set only  $\approx 55\%$  complete. We re-audited confirmatorily with *three* independent judge families reading every golden-unmatched finding against the diff (“does this identify a genuine problem?”), plus a calibration pass on golden-*matched* findings. Llama 3.3 calls 76% of unmatched findings real (calibration on known defects: 92%); Mistral Large 3 675B calls 73% (90%); but DeepSeek V3.2 calls only 33% (71%)—and inter-judge agreement on this question is poor ( $\kappa=0.23\text{--}0.58$ ), in sharp contrast to the pair judge’s  $\kappa=0.95$ . The split is definitional, not capability: DeepSeek is a strict under-caller that rejects even  $\sim 29\%$  of *known injected defects* (largely rule/style violations it declines to call genuine problems), while the two lenient judges accept warranted concerns. Three conclusions are robust under every definition: the golden set is materially incomplete, so golden-referenced precision understates all arms; the  $\geq 2$ -family set’s effective precision is bracketed at 86–95%; and near-miss findings are overwhelmingly real. The methodological lesson generalizes: same-issue *matching* is objective and judge-invariant, while “is this a real bug?” is not a question an LLM judge of any size settles—magnitude awaits human adjudication under a fixed definition.

## VII. THREATS TO VALIDITY

**Construct validity.** The industrial case study lacks authoritative ground truth, so our signals corroborate rather than prove correctness, and the LLM judge may share biases with the reviewer models. We mitigate this by triangulating four independent signals of different kinds (peer agreement, static analysis, an independent judge, and later fixes), by treating the Evidence Score only as a supporting heuristic, and by never reporting benchmark correctness for that dataset. On SWE-PRBench, matching metrics measure agreement with reviewers rather than absolute correctness, and we label them as such.

**Construct: golden incompleteness (quantified).** Even the objective benchmark is incomplete: three independent judge families agree the golden set misses real defects but disagree on how many (33–76% of unmatched findings judged real; inter-judge  $\kappa=0.23\text{--}0.58$ )—a definitional disagreement about whether rule/style violations count as genuine problems, not a capability gap (Section VI-I). Golden-referenced precision is therefore a lower bound. Relative comparisons between arms are unaffected, because incompleteness applies to every arm identically; human adjudication of a sample under a pre-fixed definition is planned.

**Construct: judge and verifier behavior.** The matching judge is effectively binary and threshold-insensitive across 0.5–0.9, which removes a hyperparameter-sensitivity concern; the verifier and matching judge are different model families from the generator (a non-circular triangle), mitigating the self-preference bias documented in prior work [11]. The cross-family pair judge is additionally validated instrument-side—two independent families agree at  $\kappa=0.95$  over all 8,061 pairs and reproduce the headline result—but its human validation (a dual-annotator  $\kappa$  on a 50-pair sample) is still pending, so the corroboration numbers remain instrument-relative until then. Cluster representatives default to the Haiku member (deterministic and disclosed); a representative-rotation sensitivity check is planned.

**Internal validity.** Differences in outcome could stem from confounds other than topology. We hold the model, prompts, specialist roles, and sampling budget fixed and review a byte-identical immutable snapshot per PR across all four architectures, so topology is the only manipulated variable; the compute-matched Generalists-3 control further separates communication structure from raw sampling. The deterministic evaluation engine and stored reproducibility metadata let us regenerate every metric from the archived run and judge caches without new model calls; those caches ( $\sim 94$  MB) are held in access-controlled storage and released via the OSF data component at publication, rather than committed to the source tree.

**External validity.** Qodo is a synthetic-injection benchmark: defects are planted (rule violations and functional bugs) into real merged PRs, and injected defects are likely cleaner and more localizable than natural review targets, so absolute scores sit on a favorable setting even though within-benchmark relative comparisons hold. The SWE-PRBench replication



on natural human review comments—which reproduces the recall-up/F1-tie pattern—partly offsets this, as does the multi-language spread of the injected corpus; the industrial arm and execution-grounded benchmarks remain future work.

**Conclusion validity.** The headline claims (the specialization null, Agentless F1 dominance, H-verify’s failure, and both H-hetero-precision sub-claims) are confirmatory: registered sample sizes, paired non-parametric tests, effect sizes with CIs, and Holm correction within families; the full secondary family will be jointly corrected once every registered secondary is computed. The finer stratifications (defect-type, false-positive anatomy, the completeness magnitudes) are exploratory and labeled as such. Three registered pieces remain pending: the RQ3 complexity-interaction analysis, the Sonnet 4.5 robustness arm at confirmatory scale, and the industrial-portal arm.

## VIII. CONCLUSION

This paper asked whether multi-agent communication topology improves automated code review. Under a pre-registered, compute-matched, frozen-model design, the answer on the *generation* axis is no: role specialization adds nothing over plain sampling (the primary null), no homogeneous topology approaches the single-pass baseline on F1, extra agents buy recall at a 3–9 $\times$  cost premium, and the self-consistency rescue suggested by our own pilot does not replicate confirmatorily. The *verification* axis is where multi-agent value lives: agreement between independent model families is a strong, registered, judge-invariant precision instrument—findings two families corroborate are real 72% of the time, findings all three agree on 89%, versus 54% for a single model repeating itself—and the signal concentrates on functional bugs, the class reviewers care most about. These results replicate concurrent general findings that teams hold experts back [13] and that mixing models does not beat the best single model on quality [14], and they provide the first pre-registered, statistically powered test of the correlated-error conjecture for AI code review [15]: cross-family agreement tracks external ground truth even without executable specifications. The practical guidance is direct: do not add homogeneous agents for quality; ship cross-family agreement as a cheap, model-agnostic trust signal on top of any single reviewer. Future work: human  $\kappa$  validation of the pair judge, fixed-definition human adjudication of benchmark completeness, the industrial-portal arm, the complexity interaction, and replication on execution-grounded review benchmarks such as c-CRAB [16]. Two exploratory probes sharpen the tooling boundary: a deterministic checker mirroring the repositories’ own lint rules recovers mechanical conventions the LLM misses (a missing `Async` suffix at 43% vs. the LLM’s 7%), so mechanical conventions belong to deterministic tooling rather than more agents; and supplying whole changed files plus one-hop dependencies at the review commit does not raise recall over the diff alone (0.37 $\rightarrow$ 0.35, n.s.), so passive context is not the lever—adaptive, agentic retrieval and execution is the frontier direction [4].

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